Mr Dimelow

11 Information handling, modelling and forward planning

11.3 - The importance, implications and ways of protecting the intellectual property rights of designers, inventors and companies

Topic	What students need to learn:
11 Information handling, modelling and forward planning	11.3 - The importance, implications and ways of protecting the intellectual property rights of designers, inventors and companies: *a) patents* *b) copyrights* *c) design rights* *d) trademarks*

Intellectual Property Rights



- **Intellectual Property (IP)** is an idea that is turned into physical creation that did not already exist *(e.g. a book, a film, camera lens mounting system, handbag design or a car name badge).*
- Ideas alone do not count as IP – a physical format of some kind is compulsory.
- Uniqueness is essential when seeking to give IP the legal protection that is afforded by **intellectual property rights (IPR).**

Intellectual Property Rights

- Some rights are ***automatically*** endowed when work is created **(copyright)**; others have to be ***applied for*** **(a patent)**.
- IP laws vary from country to country and there are various rules governing applications.

See Video:

https://www.focuselearning.co.uk/programmes/?programme=ipr&page=introduction

Intellectual Property and Intellectual Property Rights

(AGS Logins > AGS Useful Software List > Design Engineering > Focus On Series)

Patents

- The government's **Intellectual Property Office (IPO)** grants legal patent rights for novel inventions.
- Patents apply only to working parts of a design *(e.g. a car suspension system, electronic product printed circuit board (PCB), automatic kettle sensor).*
- They provide up to 20 years' legal protection but are expensive and complicated to obtain.
- It is essential that the invention is completely new or significantly improves an existing design.

Patents

- Detailed, labelled diagrams and clear descriptions are an important part of patent applications.
- A patent attorney is often employed to check a patent prior to submission.
- Legal conflict examples include *Apple vs Samsung (software features) and Dyson vs Hoover (dual cyclone).*
- It can be argued that publication of a patent encourages further innovation.

See Video: https://www.focuselearning.co.uk/programmes/?programme=ipr&page=introduction

Patents

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Copyright

- ***Copyright*** is a law that gives the owner of a work (for example, a book, movie, picture, song or website) the right to say how other people can use it.
- It allows management of permission for others to use work (e.g. licensing photos for use in books).
- The ***UK Copyright, Designs and Patents Act 1988*** provides cover for most works for 70 years after the death of the creator.



Copyright symbol

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Copy Rights

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Design Rights

Unregistered Design Rights

- **Unregistered Design Rights** provide legal protection for designers to stop unauthorised copying for ten years.
- They cover appearance (shape, ornamentation), but NOT how products work.
- Registration is NOT required, but ownership is hard to prove in a dispute.

Design Rights

Unregistered Design Rights

- Using the **copyright symbol (©)** and *metadata and watermarks on digital files* makes it easier to assert ownership.
- Early drafts, sketches and recordings, as well as deliberate, unique modifications can be used as evidence.
- The UK Copyright Service provides an independent evidence of authorship facility.

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Unregistered Design Rights

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Design Rights

Registered Design Rights

- **Registered Designs** offer legal protection for product appearance and decoration, NOT working parts.
- Designs are registered for up to 25 years by submitting illustrations and fee to the IPO.
- Trade in fake designer label products is an ongoing problem despite legal protection.

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Registered Design Rights

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Trademarks

- ***Trademarks*** give legal protection for brand identity and facilitate marketing.
- A recognisable combination of words, sounds, colours and logos is important for companies.
- Logos are instantly identifiable, unique graphical symbols used to associate products and manufacturers.
- A trademark may incorporate graphics but can also just be a word or a phrase such as ***'dual cyclone'.***

Trademarks

- Trade marks are registered by making an application and paying a fee to the IPO.
- Trademarks need to be renewed every 10 years.
- The ***trademark symbol*** is used to warn others that it is a protected trademark.



Trademark symbol

See Video: https://www.focuselearning.co.uk/programmes/?programme=ipr&page=introduction

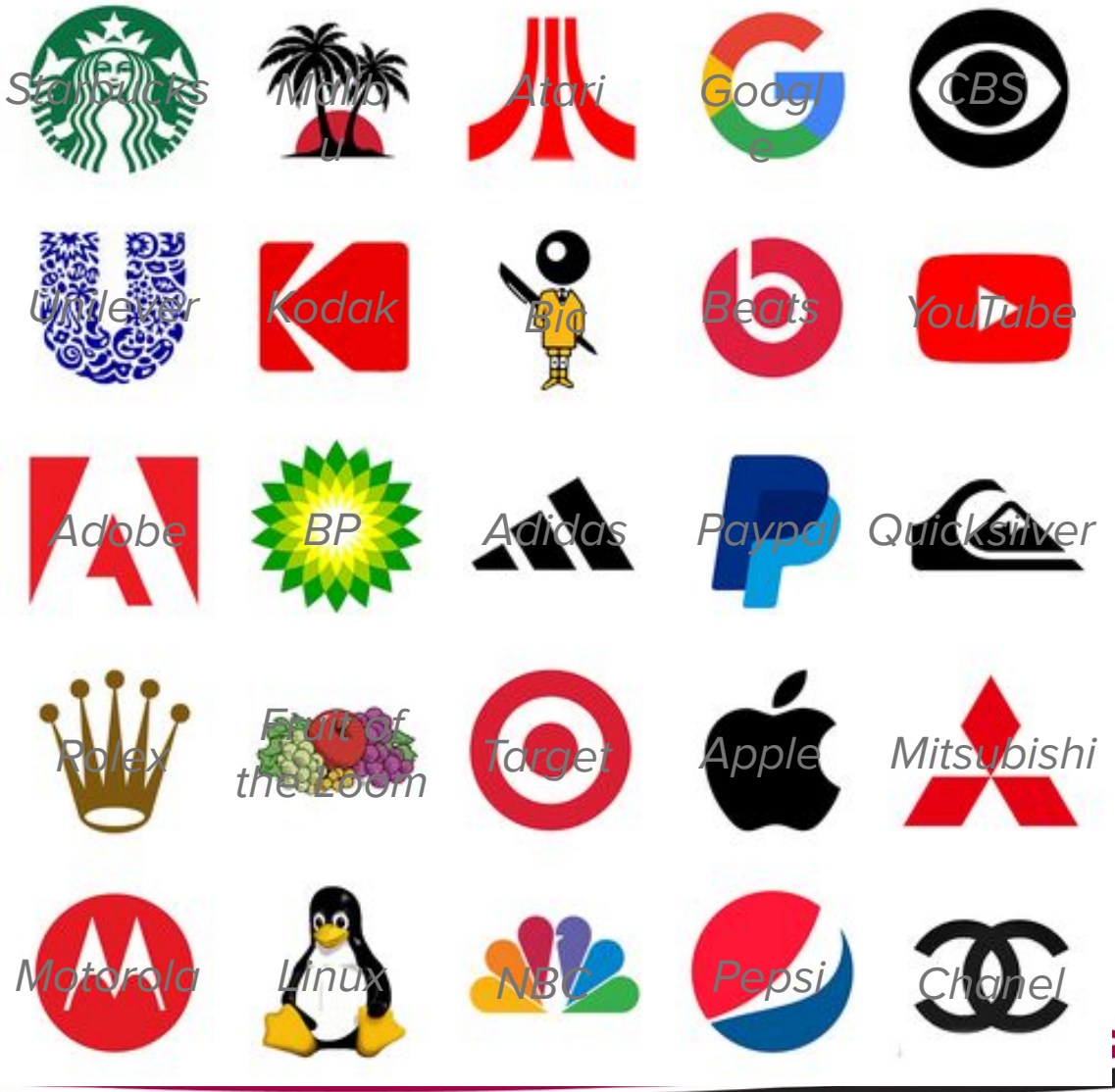
Trade Marks

(AGS Logins > AGS Useful Software List > Design Engineering > Focus On Series)

Can you identify ALL twenty five of these Logos?



Can you identify <u>ALL</u> twenty five of these Logos?



Typical Mistakes

If an exam question asks you to explain how designers protect their designs from being copied, you are expected to demonstrate your knowledge of a range of IPR.

Don’t just write about ways of keeping work secret, as this is only one aspect of the topic, which mainly affects patent applications

Exam Tip



Many products encompass **all four** of the main types of IPR, so use a specific product example that you are familiar with, such as the Bic ® Atlantis ® pens – to illustrate how designers can protect their work from being copied.

- ***Patent:*** *incorporates licensed Laszlo Biro ball ink delivery system.*
- ***Registered design:*** *the 'S' curve, barrel shape and other aesthetic features.*
- ***Trademark:*** *the Bic ® and Atlantis ® names are both registered.*
- ***Copyright:*** *all text on the packaging, such as safety instructions.*

Task

Many products encompass **all four** of the main types of IPR, so use a specific product example that you are familiar with:

Your task is to research a product you are familiar with – try and find out what intellectual property rights are used to protect it.

Possible examples:

- *Football boots, running shoes*
- *Tennis, squash, badminton rackets*
- *Mobile phones, games consoles*

Now Test Yourself

1. *What do you understand by the term 'intellectual property rights (IPR)?*

2. *Explain how IPR are used to protect products. Include at least one specific product example that covers the full range.*

3. *What are the potential difficulties that might be encountered if you rely on IPR through unregistered design rights, and how might you guard against these?*